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Enhancing semi-continuous anaerobic digestion efficiency with nanobubble water: thermophysical properties optimization and microbial regulation

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ABSTRACT

The utilization of nanobubbles to enhance biomass anaerobic digestion (AD) for bioenergy production has emerged as a prominent research focus. However, the mechanisms governing their effects on semi-continuous AD characteristics and heat-mass transfer processes remain inadequately elucidated. This research employed corn stover as feedstock for semi-continuous AD at 35 ± 1 °C, investigating the impacts of nanobubble water (NBW) prepared with H_2, CH_4, and CO_2 on digestion performance, thermophysical properties (thermal conductivity, specific heat capacity, density, viscosity, surface tension), and microbial dynamics. The results demonstrated that NBW-supplemented systems achieved up to 15.67 ± 1.96 % higher daily methane production compared to the control. Compared to the control group, NBW-supplemented systems demonstrated improvements of 7.22–29.50 % and 5.06–18.85 % in total solids and volatile solids removal efficiencies, respectively. Concurrently, soluble chemical oxygen demand and reducing sugar concentrations decreased by 7.36–13.07 % and 3.34–10.61 % in NBW groups. NBW enhanced mass transfer efficiency, thereby inducing selective microbial enrichment and functional augmentation that improved substrate utilization. Furthermore, NBW optimized digestate thermal stability and flow properties, consequently enhancing heat-mass transfer efficiency during AD. This work elucidates nanobubble-mediated enhancement mechanisms in semi-continuous AD through integrated thermophysical and microbiological perspectives, thus providing a theoretical foundation for advanced biogas production technologies.

1. Introduction

Anaerobic digestion (AD) serves as a critical technology for valorizing organic wastes, holding substantial practical significance in renewable energy development and environmental protection. The AD process fundamentally relies on microbial metabolic systems, through which complex organic matter is biochemically converted into value-added products, including methane and short-chain fatty acids [1]. However, conventional AD occurs within complex fluid systems where low mass transfer efficiency and slow reaction kinetics persist as pervasive challenges [2].

Nanobubbles (NBs) demonstrate significant potential in AD applications due to their distinctive physicochemical properties, including large specific surface area, negative zeta potential, radical generation capacity, and enhanced mass transfer rates [3]. The extremely small dimensions of NBs confer exceptional longevity in aqueous environments, thereby prolonging gas-liquid interfacial mass transfer processes and achieving superior mass transfer efficiency. For instance, CO_2 NBs exhibited 11-fold higher mass transfer coefficients and 7-fold greater buffering capacity compared to conventional CO_2 macrobubbles [4]. Numerous studies confirm that NBs enhance microbial and enzymatic activities while promoting substrate degradation, consequently improving AD efficiency [5]. However, NBs have a limited lifespan, and their effectiveness may gradually decline during long-term AD processes. Furthermore, existing studies on NB-enhanced AD have predominantly been conducted in batch-fed reactor systems, with limited

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research carried out under semi-continuous operation [6,7]. In semi-continuous systems, frequent feeding and discharge operations, along with variations in operational cycles, may influence the actual performance of NBs. Therefore, incorporating NBs into semi-continuous AD processes is of significant importance for advancing their engineering application in the field of AD. H_2 and CO_2 act as key substrates for hydrogenotrophic methanogens and play essential roles in the metabolic pathways of AD systems [8]. Meanwhile, H_2, CO_2, and CH_4 are the primary components of biogas. A systematic investigation of these three gases will also provide valuable insights for the subsequent development of biogas-nanobubble water (biogas-NBW).

In AD systems, the thermophysical properties of digestate directly influence heat and mass transfer efficiencies within reactors. The thermophysical properties of digestate mainly include density, viscosity, surface tension, thermal conductivity, and specific heat capacity, which are influenced by the composition of the substance, substrate concentration, temperature, microbial metabolites, and other aspects [9,10]. Density, viscosity, and surface tension collectively determine rheological behavior, thereby affecting mass transfer rates. For instance, elevated substrate concentrations increase digestate viscosity, impairing fluidity [11]. Reduced surface tension enhances molecular diffusion, consequently improving mass transfer efficiency. Critically, diminished surface tension correlates with accelerated organic degradation and biogas production [12]. Thermal conductivity and specific heat capacity define heat transfer characteristics, jointly regulating thermal energy transport and temperature distribution. Studies have confirmed that higher total solids (TS) concentrations decrease specific heat capacity in pulp substrates [13]. Compared to pure water, the total heat transfer coefficient of the reactor decreased by 49.51 % when the digestate was used as a fluid [14]. Therefore, the thermophysical properties of the digestate can directly affect the matter and energy transfer in the system. The poor thermophysical properties increase the resistance to thermal mass transfer and impact the growth and metabolism of microorganisms, which is the current challenge of AD.

The superior gas-liquid mass transfer efficiency of NBs inherently induces thermophysical modifications in digestate. Current research predominantly focuses on NBs impacts on AD end-products, while systematic investigations into their effects on digestate thermophysical properties remain underexplored. Some studies have currently explored the physicochemical properties of single liquid-phase systems of NBW through simulations. For example, when different dimensions and numbers of NBs burst, there were different density and temperature variations in their regions [15]. In addition, some calculations found that the presence of N_2, CO_2, and O_2 NBs increased the specific heat capacity of dodecane and isooctane [16]. However, the influence regularities of NBW on the thermophysical properties of the complex liquid-phase system of digestate needs to be supplemented.

In this research, the effects of NBW with different gas species on the properties of corn stalk semi-continuous AD and the thermophysical properties of digestate were investigated experimentally. The research specifically examines: (1) impacts of various NBW types on methane production and substrate degradation efficiency; (2) modifications of digestate density, viscosity, surface tension, thermal conductivity, and specific heat capacity during semi-continuous operation; (3) microbial community restructuring and functional gene expression alterations induced by NBW supplementation. The findings provide critical technical parameters for designing and optimizing NBW-coupled AD systems, advancing technological upgrading of high-efficiency biogas production platforms.

2. Methods

2.1. Preparation of raw materials

The experiment used corn stalks as the substrate for digestion and activated sludge as the inoculum. The raw material characteristics are shown in Table 1. Corn stalks were obtained from the Xuchang experimental field of Henan Agricultural University, crushed and sieved through a 60-mesh sieve. The activated sludge was collected from the Zhengzhou Matougang wastewater treatment plant and activated for two weeks at 35 °C in an incubator before use. A cavitation nanobubble generator (ZJC-NM-200L, ZJ Environmental Protection, China) was used to prepare NBW with a gas flow rate of 50 mL/min and a NBW flow rate of 3 L/min, employing a circular method (volume 2 L, duration 5 min). The experiments employed the three primary components of biogas (H_2, CH_4, and CO_2) as carrier gases, producing H_2-NBW, CH_4-NBW, and CO_2-NBW, respectively.

2.2. Experimental protocol and device

The experiment employed a continuous stirred tank reactor (CSTR) with a total volume of 4.4 L and a working volume of 3 L (Fig. 1). The feedstock was introduced into the reactor through the feed inlet and underwent AD. The digested effluent was discharged from the effluent outlet, while the residual sludge was expelled from the sludge discharge port located at the bottom of the reactor. The four equally sized CSTR AD reactors were named A1, A2, A3, and A4. A1-A4 used deionized water, H_2-NBW, CH_4-NBW, and CO_2-NBW, respectively. AD was conducted in semi-continuous operation with 14 days start-up period. During the start-up phase, NBW would be mixed with sludge and corn stalks before being poured into the reactor for anaerobic digestion. Started with small amounts of substrate and began semi-continuous operation after the methane concentration stabilized. During the semi-continuous operation phase, after daily efflux, an equal volume of NBW and substrate mixture was added to the reactor. The hydraulic retention time (HRT) and sludge retention time (SRT) for semi-continuous operation were both 15 days, and the volumetric loading rate (VLR) was 2 gVS/(L·d). The stirring frequency was 6 times/day, 15 min/time.

2.3. Analytical instruments and methods

Methane concentration was determined using a gas chromatograph (6820 GC, Agilent, USA) equipped with a 5A molecular sieve packed column and a thermal conductivity detector (TCD). Nitrogen was used as the carrier gas at a flow rate of 45 mL/min. The oven temperature was set at 80 °C, while the detector and injector temperatures were maintained at 150 °C and 100 °C, respectively. The pH value was measured with a pH meter (PHS-3E, Leici, China). Total solids (TS) content was determined by drying the samples in an oven (101, Zhongxing, China) at 105 °C until a constant weight was achieved. Volatile solids (VS) content was measured by combusting dried samples in a muffle furnace (MF-2200, Chuangxin Instrument, China) at 600 °C for 3 h. The volatile fatty acid (VFA) concentration was analyzed using a gas chromatograph (7890B, Agilent, USA) equipped with a DB-FFAP column and a flame ionization detector (FID). Nitrogen was employed as the carrier gas at a flow rate of 30 mL/min. The column temperature was set at 230 °C, and both the injector and detector temperatures were maintained at 250 °C. Reducing sugars were determined via the anthrone colorimetric method, and soluble chemical oxygen demand (SCOD) was measured by the potassium dichromate spectrophotometric method [17,18].

The density of the digestate was determined using a density meter

Table 1
Raw material characteristic parameters.

Parameters	Corn stalk	Activated sludge
TS (%)	92.72 ± 0.10	5.95 ± 0.02
VS (%)	80.55 ± 0.21	2.17 ± 0.02
C (%)	38.18 ± 0.02	0.41 ± 0.01
H (%)	5.81 ± 0.02	11.14 ± 0.01
N (%)	1.27 ± 0.01	0.09 ± 0.01
C/N	30.06 ± 0.34	4.56 ± 0.73

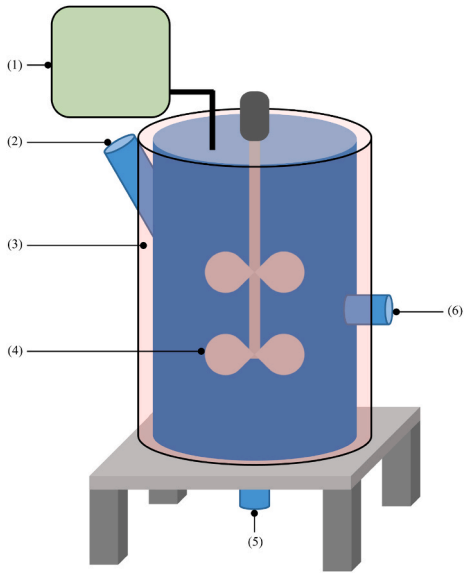


Fig. 1. Anaerobic digestion reactor: (1) gas collection bag, (2) feed inlet, (3) thermostatic layer, (4) stirrer, (5) sludge discharge port, (6) effluent outlet.

(XF-120YT, Lichen, China) based on the Archimedes buoyancy principle. Apparent viscosity was measured using a rotational viscometer (NDJ-5S, Shanghai Hengping, China) equipped with a #0 rotor at rotational speeds of 6, 12, 30, and 60 rpm. Thermal conductivity was measured via the transient plane source method using a thermal conductivity analyzer (HNB-DRS2, Xiamen Senbei, China) [19]. Surface tension was characterized by a contact angle measuring instrument (Alpha-s, Guangzhou Beto, China) employing the pendant drop method [20]. Specific heat capacity was measured using a simultaneous thermal analyzer (SDT650, TA Instruments, USA) under a nitrogen atmosphere with a flow rate of 50 mL/min, and the temperature was increased from 30 °C to 50 °C at a heating rate of 1.5 °C/min. The nuclear magnetic resonance spin-spin relaxation time (T_2) was analyzed using a low-field nuclear magnetic resonance imaging system (MesoMR23-060H-I, Niumag, China). All measurements were performed in triplicate to minimize experimental error.

2.4. Microbiological analyses

Microbial samples were collected from the original microorganisms (OM) and terminal microbial communities of semi-continuous AD groups (CG: control group; H-NBW: H_2-NBW; CH_4-NBW; M-NBW; C-NBW: CO_2-NBW). Microbial community structure and functional genes were detected using high-throughput sequencing by Sangon Biotech Co., Ltd. (Shanghai, China). DNA extraction was performed using the E.Z.N. A™ Mag-Bind Soil DNA Kit (Omega, M5635-02, USA). Bacterial amplification targeted the V3-V4 hypervariable regions using primers 341F/805R. Archaeal communities were analyzed through nested PCR amplification. The first-round amplification was performed using primers 340F and 1000R, followed by secondary amplification with primers 349F and 806R, targeting the V3-V4 hypervariable regions of the 16S rRNA gene. Library quality was verified by 2 % agarose gel electrophoresis prior to sequencing on the Illumina NextSeq 2000 platform (Illumina, USA). PICRUSt (v1.1.4) and FAPROTAX (v1.2.1) were used for annotation of 16S rRNA gene sequencing data to predict microbial metabolic functions.

2.5. Energy conversion efficiency analysis

The energy conversion efficiency assessment was calculated using the following equation:

$$W_e = P \cdot 60 / Q \quad \text{(Eq. 1)}$$

where W_e is the electrical energy consumed in preparing 1 L of NBW (J/L), P is the power of the nanobubble generator (380 W), and Q is the NBW flow rate (3 L/min).

$$W_{cir} = W_e \cdot (Q \cdot t_{cir} / V_{cir}) \quad \text{(Eq. 2)}$$

where W_{cir} is the energy consumed in the circulation preparation of 1 L of NBW (J/L), t_{cir} is the circulation time (5 min), and V_{cir} is the circulation volume (2 L).

$$V_n = V_o(1 - R_{inoc}) / V_e \quad \text{(Eq. 3)}$$

where V_n is the volume of NBW required per unit volume of the reactor (L/L), V_o is the output volume (0.20 L), R_{inoc} is the inoculum ratio (0.30), and V_e is the effective volume of the reactor (3 L).

$$W_{ctrl} = Q_{CH_4} \cdot V_{ctrl} / V_m \quad \text{(Eq. 4)}$$

where W_{ctrl} is the energy output per unit volume for the control group (kJ/L), Q_{CH_4} is the lower heating value of methane (803.49 kJ/mol), V_{ctrl} is the volumetric methane production of the control group (L/L), and V_m is the molar volume of the gas (22.40 mol/L).

$$W_{NBW} = W_{ctrl} \cdot r \quad \text{(Eq. 5)}$$

where W_{NBW} is the net energy output of the NBW group compared to the control group (kJ/L), and r is the methane production enhancement rate of the NBW group (dimensionless).

$$\eta = W_{NBW} \cdot 1000 / W_{cir} \cdot V_n \quad \text{(Eq. 6)}$$

where η is the net energy ratio representing the gain of the NBW group compared to the control group (dimensionless).

2.6. Data processing and analyses

All experimental datasets were processed and statistically analyzed using Microsoft Excel (v2016) for data organization and one-way ANOVA testing, with a statistical significance threshold established at $p = 0.05$. Origin (v2022) was employed for graphical analysis, where all data visualizations incorporated error bars representing the standard deviation to illustrate data dispersion.

3. Results and discussion

3.1. Methane production

Daily methane yield first increased and then stabilized over time, achieving stability during the second HRT (Fig. 2a). Throughout the semi-continuous operation period, NBW-supplemented groups exhibited significantly higher methane yields than the control group ($p < 0.05$). During stable methanogenic operation (HRT 2–3), the NBW groups demonstrated 9.68 ± 2.08 % (H_2-NBW), 7.34 ± 2.07 % (CH_4-NBW), and 15.67 ± 1.96 % (CO_2-NBW) increases in daily methane production compared to the control group (159.88 ± 2.34 mL/(gVS·d)). Cumulative methane yield in NBW groups increased by 11.71 ± 2.50 % (H_2-NBW), 9.92 ± 2.42 % (CH_4-NBW), and 16.08 ± 2.45 % (CO_2-NBW) compared to the control (37.90 ± 0.61 L total). Although different NBs exerted distinct effects on substrate degradation and microbial community composition, all demonstrated positive impacts on methanogenesis [21]. Notably, CO_2-NBW achieved the highest daily methane yield and cumulative yield in semi-continuous AD compared to H_2-NBW and CH_4-NBW. The differential effects of identical gas-species NBW may arise from variations in bubble generation conditions and AD environments. For instance, H_2-NBW demonstrated superior methanogenic enhancement in swine manure AD systems, whereas CO_2-NBW showed

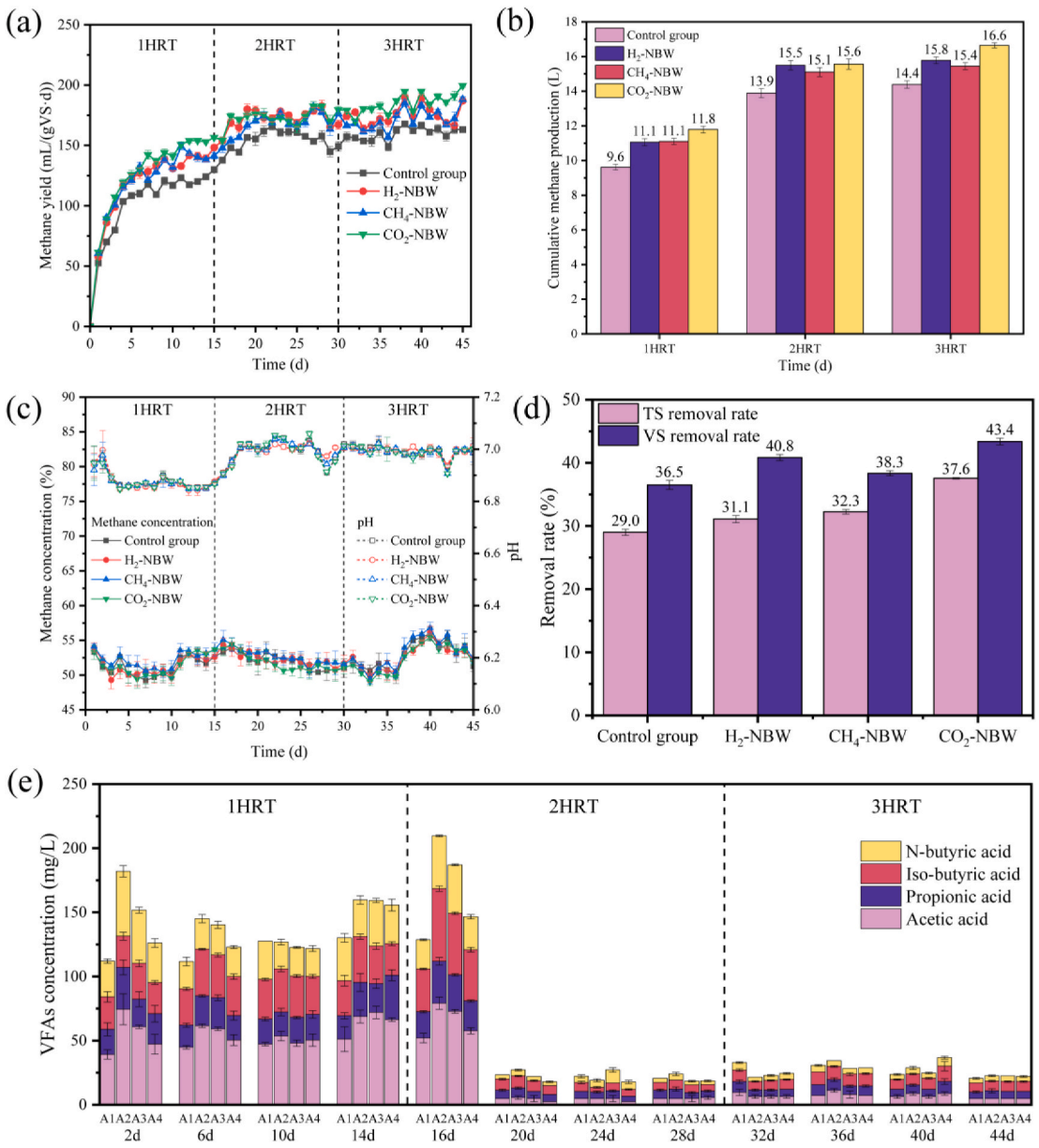


Fig. 2. (a) Methane yield dynamics, (b) cumulative methane production per HRT, (c) methane concentration and pH profiles, (d) TS and VS removal rates, (e) VFAs concentration (A1: control, A2: H_2-NBW, A3: CH_4-NBW, A4: CO_2-NBW).

greater methane production performance in semi-continuous AD of rice straw substrates [22,23]. The superior performance of CO_2-NBW in semi-continuous systems may relate to enhanced gas dissolution kinetics under continuous liquid-phase mixing.

The stimulatory effects of NBs gradually attenuated during prolonged operation (Fig. 2b). During the first HRT, cumulative methane production increased by 14.95 ± 2.86 % (H_2-NBW), 15.45 ± 2.66 % (CH_4-NBW), and 22.56 ± 2.86 % (CO_2-NBW) compared to controls. These enhancements declined to 11.57 ± 2.84 %, 8.76 ± 2.71 %, and 12.02 ± 2.94 % in the second HRT, further reducing to 9.68 ± 2.08 %, 7.34 ± 2.07 %, and 15.67 ± 1.96 % in the third HRT. This deterioration likely resulted from reduced NBs persistence under continuous stirring and feeding-induced hydraulic shear forces. Lower pH conditions and elevated organic matter concentrations were found to significantly reduce NBs longevity, with persistence durations typically ranging between several days and weeks under such operational parameters [24]. Notably, methane concentration remained statistically indistinguishable between NBW groups and controls ($p > 0.05$, Fig. 2c). Similarly, N_2-NBW increased methane production by 29 % in sludge AD, whereas methane concentration showed no statistically significant variation [25]. This phenomenon reflects the dynamic equilibrium between methanogenic and acidogenic rates. As demonstrated in previous research, NBs facilitated the disruption of oligosaccharide layers on cellulose surfaces, thereby enhancing the hydrolysis of lignocellulosic substrates [5]. However, the increased substrate availability concurrently elevates both biogas production and competitive gas generation (*e.g.*, CO_2/H_2), ultimately maintaining stable methane concentration (Fig. S1).

3.2. Substrate degradation

3.2.1. pH, VFAs concentration and TS/VS removal rates

The pH initially increased, subsequently decreased, and then gradually rose and stabilized (Fig. 2c). During system initiation, the higher pH of the fresh feedstock compared to reactor content caused transient pH elevation. Progressive substrate hydrolysis during continuous

feeding reduced system pH to 6.85 ± 0.05, coinciding with accelerated VFAs accumulation. Transition to methanogenic stability phase facilitated rapid VFAs consumption, driving pH recovery and stabilization at 7.00 ± 0.05. CO_2-NBW enhanced VFA turnover under moderate substrate loading, though its buffering capacity diminished at extremes [26]. No significant pH variation ($p > 0.05$) was observed between NBW groups and the control, likely due to hydraulic shear forces and extended operational duration attenuating NB-mediated pH regulation.

The VFAs dynamics reflected the pH variations, with total VFAs concentrations in the NBW groups exceeding those of the control during the initial accumulation phase (0–16 days) (Fig. 2e). On day 16, the total VFAs concentrations in H_2-NBW, CH_4-NBW, and CO_2-NBW increased by 62.99 ± 9.34 %, 45.44 ± 5.01 %, and 14.00 ± 5.48 %, respectively, compared to the control group (128.60 ± 4.40 mg/L). This aligns with prior findings where CO_2/Air-NBW enhanced VFAs production in cellulose digestion through improved hydrolase activity [27].

NBW supplementation significantly improved substrate conversion efficiency, with H_2-NBW, CH_4-NBW, and CO_2-NBW achieving 11.89 ± 2.56 %, 5.06 ± 2.33 %, and 18.85 ± 2.73 % increases in VS removal, and 7.22 ± 2.65 %, 11.24 ± 2.24 %, and 29.50 ± 2.22 % TS reduction versus control (Fig. 2d). Some research suggested that CO_2-NBW supplementation has been documented to enhance β-glucosidase activity, consequently improving TS and VS removal efficiencies [28]. The coupled phenomena of increased VFAs concentration and reduced solid content confirm that NBW can enhance the hydrolytic-acidogenic phase. Enhanced hydrolysis of corn stover under NBW treatment generated increased quantities of low-molecular-weight organic compounds, thereby significantly enhancing substrate utilization efficiency.

3.2.2. *SCOD and reducing sugars*

As illustrated in Fig. 3a, the SCOD exhibited an initial increase followed by stabilization, primarily governed by the dynamics of substrate concentration within the reactor. The initial SCOD elevation resulted from feed-induced stimulation of hydrolytic microbial activity, thereby enhancing the release of soluble organic matter. Subsequently, a kinetic equilibrium was established among hydrolysis, acidogenesis, and methanogenesis processes. Hydrolysis products were sequentially converted to VFAs, which were rapidly consumed for methanogenesis,

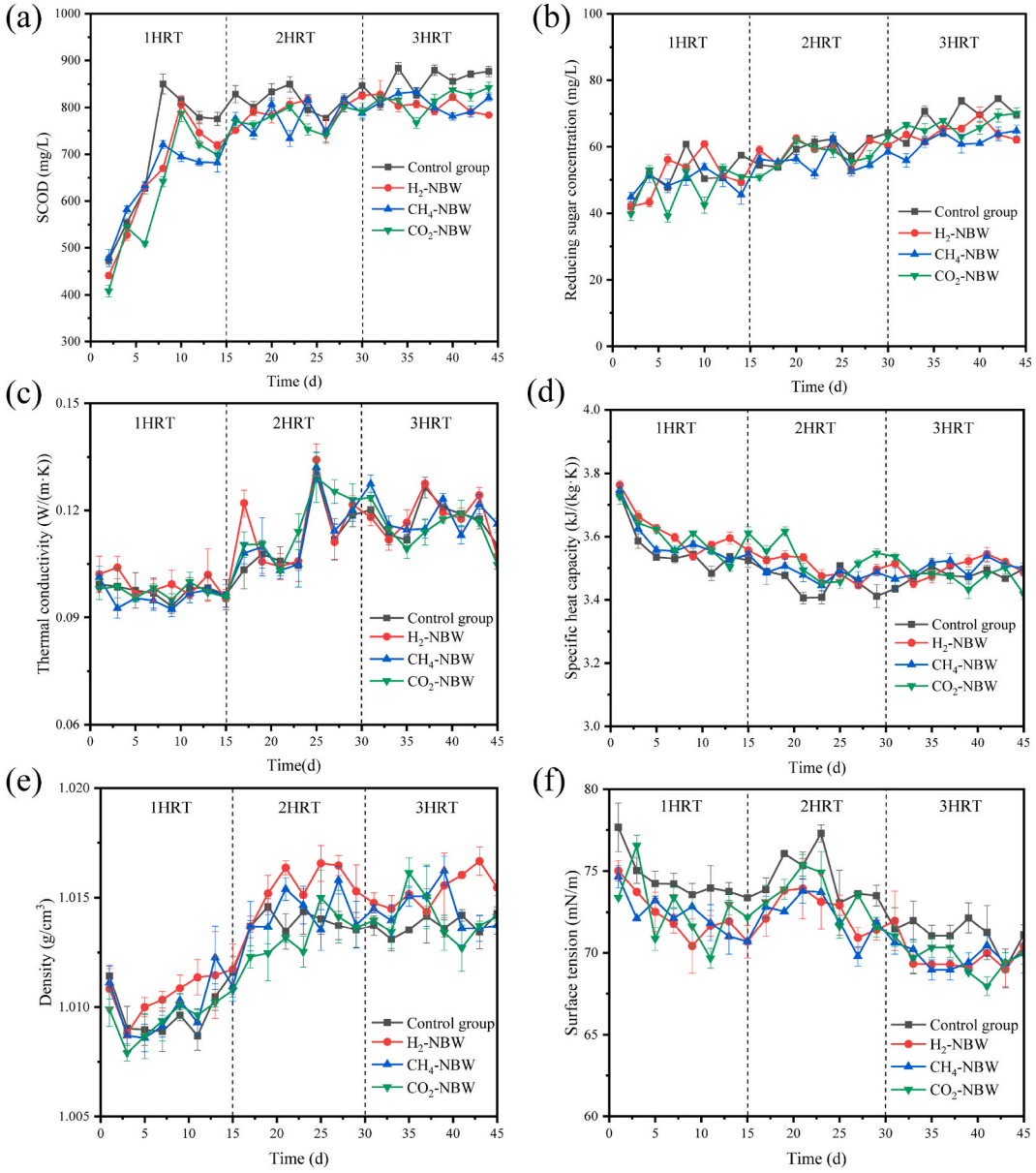


Fig. 3. (a) SCOD variation, (b) reducing sugar concentration, (c) thermal conductivity, (d) specific heat capacity, (e) density, (f) surface tension.

maintaining constant organic matter levels. Notably, NBW-supplemented systems demonstrated 7.36–13.07 % lower SCOD concentrations than the control during the first HRT. This indicates enhanced utilization of soluble substrates for microbial metabolism and growth. Similarly, the implementation of Air-NBs for wastewater aeration demonstrated a substantial enhancement in COD removal efficiency, increasing from 41.21 % (conventional bubble aeration) to 92.5 % [29].

Reducing sugars, as primary hydrolysis products in lignocellulosic digestion, reflected substrate hydrolysis-acidification progression (Fig. 3b). Initial accumulation occurred when hydrolysis rates exceeded acidogenic consumption. System stabilization established a metabolic equilibrium where sugar generation and consumption rates converged. Compared to the control, the reducing sugar concentration in the NBW groups exhibited a 3.34–10.61 % reduction. This indicates that NBW supplementation accelerated the reducing sugar consumption rate, facilitating enhanced conversion to VFAs through the glycolytic pathway, consistent with the VFAs concentration dynamics (Fig. 2e). These observations align with prior findings in swine manure AD, where NBW not only increased substrate hydrolysis rates but also accelerated the consumption of soluble proteins and carbohydrates [23].

3.3. Thermophysical properties of digestate

3.3.1. Heat transfer properties

The digestate constitutes a multiphase system encompassing suspended solids, soluble compounds, and biogas; these components differentially influence heat transfer characteristics based on their physicochemical properties. As illustrated in Fig. 3c, the thermal conductivity of the digestate exhibited an initial decrease, followed by an increase and subsequent stabilization during semi-continuous operation, with overall values ranging from 0.09 to 0.14 W/(m·K). The early-phase reduction (days 0–9) correlated with transient substrate accumulation following feeding, which increased solid fraction content. Subsequently, enhanced microbial degradation converted particulate organics into soluble constituents, progressively elevating thermal conductivity. While higher moisture content generally increases thermal conductivity, elevated straw content reduces it due to porous biomass structure [30]. Notably, the presence of solid particulates occasionally resulted in conductivity exceeding that of pure water [31]. One-way ANOVA revealed no statistically significant intergroup differences ($p = 0.95$), indicating minimal NBW impact on digestate thermal conductivity.

Fig. 3d illustrates the gradual decrease and subsequent stabilization of digestate specific heat capacity during semi-continuous operation. Composed primarily of water, soluble organics, straw residues, and suspended solids, these constituents collectively determine the specific heat capacity. Due to the significantly higher specific heat capacity of water (4.18 kJ/(kg·K)), an increase in substrate concentration typically reduces the overall specific heat capacity of the digestate [32]. This trend is consistent with observations relating to soybean molasses wastewater, where the specific heat capacity was found to decrease progressively as the TS concentration increased [33]. During initial reactor operation, escalating substrate concentrations were correlated with a decline in specific heat capacity. Specific heat capacity quantifies the thermal energy required to alter the temperature of a material, indicating its susceptibility to external thermal perturbations. Lower specific heat values accelerate temperature response rates and amplify thermal fluctuations. The H_2-NBW group exhibited the highest mean specific heat capacity (3.54 ± 0.07 kJ/(kg·K)), representing a 1.23 ± 0.33 % increase compared to the control ($p < 0.05$). This enhancement primarily stems from NBW-induced degradation of solid substrates into soluble fractions, which possess higher heat capacities. Thermal stability is critical for sustained AD performance. Consequently, NBW supplementation enhances the thermodynamic resilience of the system by modulating heat capacity characteristics.

3.3.2. Density and rheological properties

In AD of straw biomass, digestate density is primarily determined by substrate concentration. Overall, the density demonstrated an initial decrease followed by a gradual increase and stabilization during semi-continuous operation (Fig. 3e). During the early operational phase (days 0–3), the discharge of accumulated non-degradable solids from the startup period led to a reduction in digestate density. Subsequently, continuous feeding progressively increased organic matter concentration within the reactor, elevating density until stabilization. This agrees with observations in dairy manure digestion, where higher TS content correlates with increased digestate density [34]. NBW-supplemented groups exhibited marginally lower densities than the control during the initial 3 days, subsequently showing slightly higher values. The early-phase density reduction in NBW systems resulted from the intrinsic lower density of NBW compared to deionized water. The addition of NBW promoted the release of soluble substrates, resulting in an increase in digestate density.

Liquid surface tension reflects the magnitude of intermolecular forces, with digestate surface tension being significantly influenced by soluble organic constituents [35]. As shown in Fig. 3f, surface tension exhibited a generally decreasing trend with periodic fluctuations throughout the semi-continuous AD process. The surface tension of digestate decreases as the concentration of soluble substrates increases. This inverse relationship is evidenced by a strongly negative Spearman correlation between surface tension and VFAs concentrations under long-chain fatty acid stress conditions [36]. During the early operation (days 0–15), escalating soluble organic content progressively reduced surface tension. The mid-phase (days 15–23) showed temporary tension elevation due to rapid VFAs consumption by accelerated methanogenesis. Subsequently, the increased reducing sugars from cellulose/-hemicellulose hydrolysis caused a slight reduction in tension in the final phase. Throughout the semi-continuous operation, digestate surface tension in NBW-supplemented groups remained consistently lower than in the control. Research suggests that NBs function as Janus-like particles, semi-hydrophilic semi-hydrophobic entities that weaken interfacial tension by disrupting hydrogen-bonding networks in water, consequently reducing surface tension [37]. Concurrently, NBs stimulate the acidogenic phase, and the resultant elevation in VFAs concentration further diminishes digestate surface tension. The attenuation effect weakened during later operation due to a decrease in NBs concentration. Mean surface tension values were quantified as 71.49 ± 0.70 mN/m (H_2-NBW), 71.43 ± 0.45 mN/m (CH_4-NBW), and 71.83 ± 0.47 mN/m (CO_2-NBW), representing 2.56 ± 1.40 %, 2.64 ± 1.19 %, and 2.10 ± 1.21 % reductions, respectively, versus the control (73.37 ± 0.77 mN/m).

Digestate exhibits non-Newtonian fluid behavior, with viscosity primarily governed by substrate concentration, encompassing both soluble and insoluble fractions. As evidenced in Fig. 4, the apparent viscosity of digestate decreased with increasing rotational speed, demonstrating characteristic shear-thinning behavior consistent with documented rheological properties of anaerobic sludge [11]. During semi-continuous operation, viscosity progressed through three distinct phases: initial decline, subsequent increase, and eventual stabilization. Early-phase viscosity reduction (days 0–9) resulted from rapid microbial proliferation and discharge of startup-phase insoluble solids, coupled with accelerated consumption of soluble organics. This is in accordance with batch digestion systems, where a proportional decrease in viscosity was observed with a concomitant reduction in solids [38]. Subsequently, the substrate concentration within the digestate increased progressively until stabilization, resulting in a corresponding rise in viscosity to a steady-state value. Rheological measurements across varying rotational speeds demonstrated significantly lower apparent viscosity in NBW-supplemented groups compared to the control. This reduction primarily stems from NB-enhanced degradation of lignocellulosic substrates, whereby NBW supplementation promoted microbial hydrolysis of solid fractions into soluble constituents. The viscosity-lowering effect

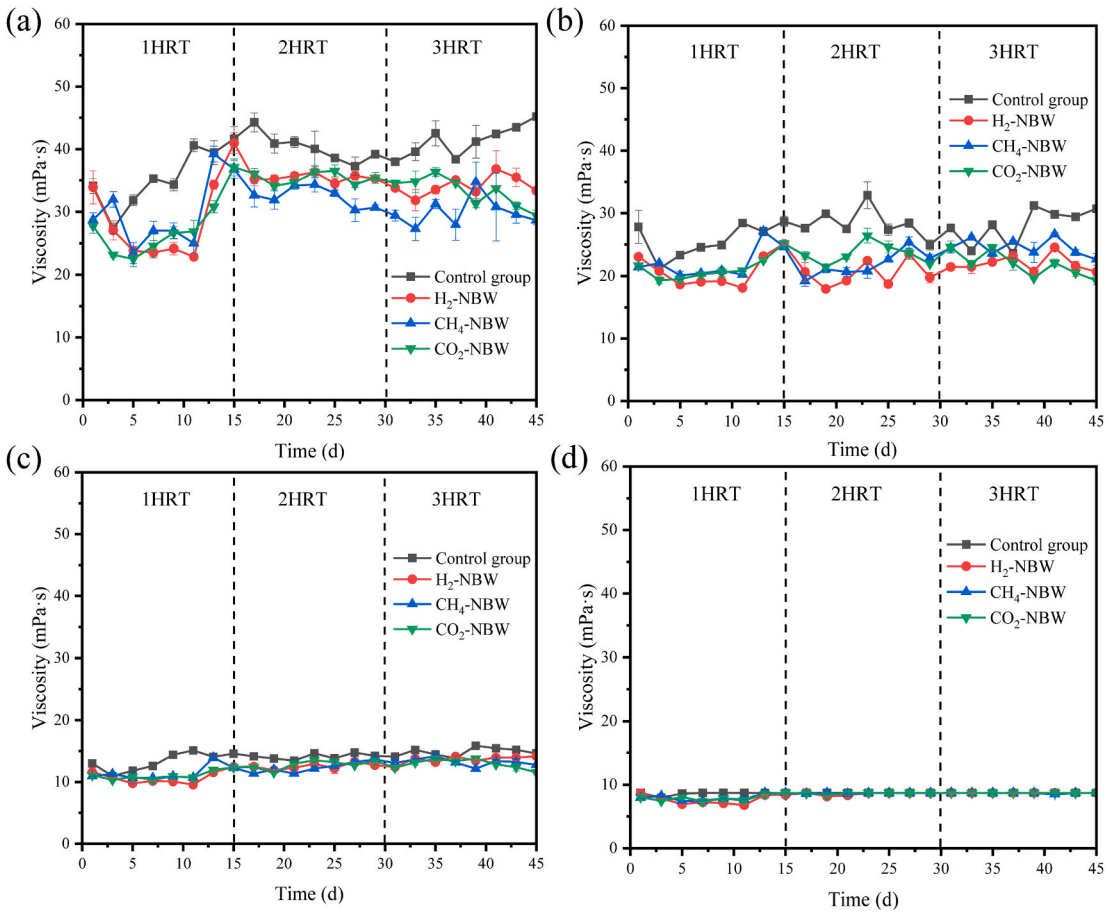


Fig. 4. Digestate viscosity at varying rotational speeds: (a) 6 rpm, (b) 12 rpm, (c) 30 rpm, (d) 60 rpm.

was pronounced during early operation but diminished progressively over time. At 60 rpm, mean viscosities during the initial 15 days measured 8.59 ± 0.01 mPa s (control), 7.63 ± 0.03 mPa s (H_2-NBW), 7.97 ± 0.06 mPa s (CH_4-NBW), and 7.90 ± 0.02 mPa s (CO_2-NBW). However, viscosity convergence occurred beyond day 15, indicating temporally constrained efficacy of NBW on digestate rheology.

3.4. Microbial community structure

AD involves synergistic microbial consortia, with distinct community structures emerging across metabolic phases. The Shannon index and Simpson index are routinely employed to quantify microbial community diversity. An elevated Shannon index signifies greater community diversity, whereas the Simpson index exhibits an inverse relationship with diversity [39]. As presented in Table 2, archaeal diversity increased substantially after long-term digestion compared to the initial inocula, whereas bacterial diversity declined. This was due to the weak methanogenic capacity of the inoculum in the early stage, and a single species of methanogens. In the later stage, with the increase of available substrate, all kinds of methanogens reproduced in large quantities, ultimately forming a stable and rich archaeal community structure. Notably, NBW-supplemented systems exhibited lower Simpson indices for both archaea and bacteria relative to controls. This reduction signifies NBW-mediated enrichment of microbial biodiversity, thereby augmenting functional redundancy within the anaerobic ecosystem.

Under fixed temperature, substrate composition, and operational conditions, AD systems exert a directional enrichment effect on microbial consortia [40]. Compared to the initial inocula, long-term operation substantially altered dominant bacterial genera across all groups (Fig. 5). The control group developed *DMER64* as the predominant genus, which is a propionate/butyrate-oxidising hydrogen producer capable of direct interspecies electron transfer (DIET) with methanogens

Table 2
Microbial community diversity indices.

Microorganisms	Sample	Number	OTUs	Shannon	Simpson	Coverage
Bacteria	OM	111898	1446	5.225834	0.015297	0.998963
	CG	78416	1702	5.199287	0.022549	0.996748
	H-NBW	75352	1714	5.313968	0.013539	0.99712
	M-NBW	93753	1772	5.259141	0.015052	0.997536
	C-NBW	81599	1744	5.148599	0.017458	0.997034
Archaea	OM	127101	48	0.543364	0.79764	0.999976
	CG	101665	114	1.946366	0.321728	0.999911
	H-NBW	128538	127	2.366265	0.171799	0.999922
	M-NBW	102531	111	2.132742	0.246835	0.999922
	C-NBW	101491	108	2.150953	0.199211	0.999911

Notes: OM: original microorganisms, CG: control group, H-NBW: H_2-NBW, M-NBW: CH_4-NBW, C-NBW: CO_2-NBW.

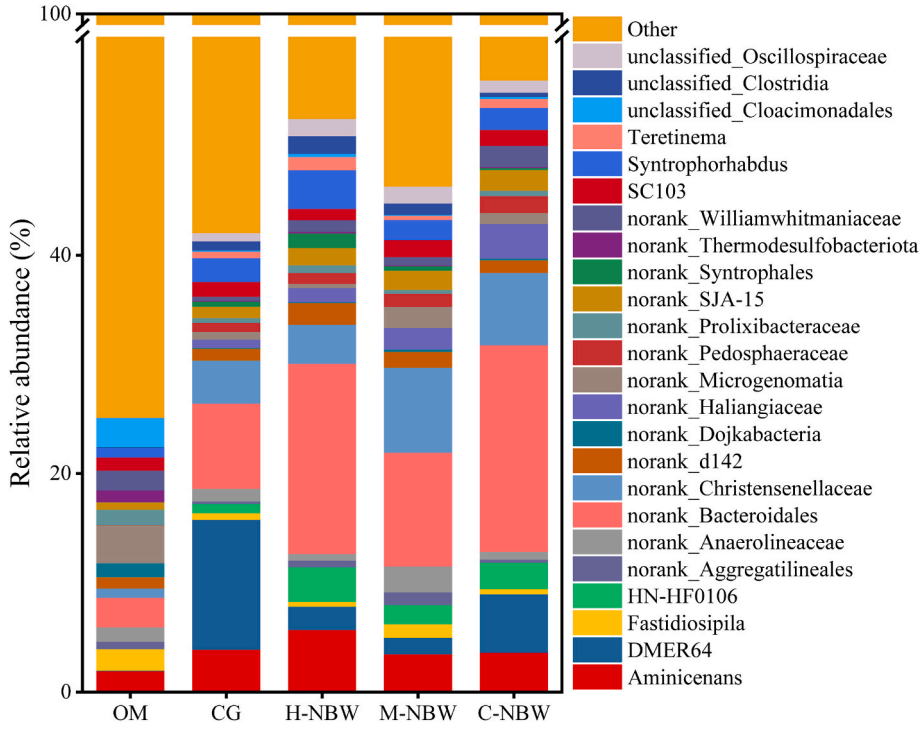


Fig. 5. Genus-level bacterial community structures (OM: original microorganisms, CG: control group, H-NBW: H_2-NBW, M-NBW: CH_4-NBW, C-NBW: CO_2-NBW).

[41]. In contrast, NBW-supplemented systems established *norank_Bacteroidales* dominance, a polysaccharide-fermenting acidogen secreting multiple glucosidases to produce acetate, propionate, and butyrate [39]. Notably, compared with the control group, the relative abundance of the genus *HN-HF0106* in the NBW group was increased by 2.7-fold (H_2-NBW), 1.1-fold (CH_4-NBW), and 1.8-fold (CO_2-NBW), respectively. *HN-HF0106* can degrade cellulose to produce H_2 and acetate, and its increased relative abundance facilitates the degradation of cellulose-based substrates [42]. Furthermore, cellulolytic/polysaccharide-degrading bacteria (*norank_SJA-15*, *norank_Dojkabacteria*), proteolytic bacteria (*norank_Haliangiaceae*, *norank_Williamwhitmaniaceae*), and acidogenic bacteria (*norank_Aggregatilineales*, *unclassified_Oscillospiraceae*) increased substantially in relative abundance. Collectively, this selective metabolic enrichment accelerated substrate

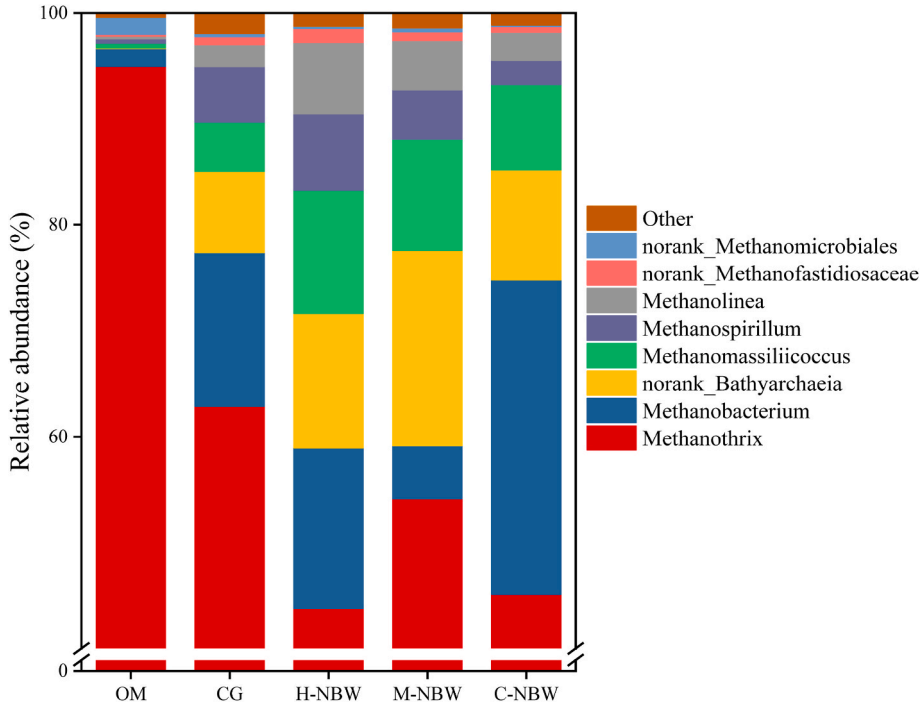


Fig. 6. Genus-level archaeal community structures (OM: original microorganisms, CG: control group, H-NBW: H_2-NBW, M-NBW: CH_4-NBW, C-NBW: CO_2-NBW).

utilization efficiency, thereby unlocking greater methanogenic potential.

The dynamic interspecific competition for substrates and ecological spaces drives continuous microbial community restructuring. At the archaeal genus level, *Methanothrix* (acetoclastic methanogen) remained dominant across all groups (Fig. 6). Compared to the original inoculum, long-term operation reduced the relative abundance of *Methanothrix* in all systems while increasing that of *Methanobacterium*, *norank_Bathyarchaeia*, *Methanomassiliicoccus*, *Methanospirillum*, and *Methanolinea*. *Norank_Bathyarchaeia* possesses lignocellulolytic capabilities and potential methylotrophic methanogenic functions [43]. *Methanobacterium*, *Methanospirillum*, and *Methanolinea* are hydrogenotrophic methanogens, while *Methanomassiliicoccus* represents a hydrogen-dependent methylotrophic methanogen [27,41,44]. NBW supplementation significantly reduced the relative abundance of the acetoclastic methanogen *Methanothrix*, with decreases of 30.41 % in H_2-NBW, 13.92 % in CH_4-NBW, and 28.26 % in CO_2-NBW systems compared to the control group. Meanwhile, the relative abundance of *norank_Bathyarchaeia*, *Methanomassiliicoccus*, and *Methanolinea* in the NBW group was higher than that in the control group. Compared with the control group, the relative abundance of *norank_Bathyarchaeia* increased by a maximum of 140.71 % (CH_4-NBW), *Methanomassiliicoccus* by a maximum of 150.46 % (H_2-NBW), and *Methanolinea* by a maximum of 227.08 % (H_2-NBW). Except for the acetoclastic methanogen *Methanothrix*, both H_2-NBW and CO_2-NBW groups were dominated by the hydrogenotrophic methanogen *Methanobacterium*, but CH_4-NBW was dominated by *norank_Bathyarchaeia*. *Methanobacterium* uses H_2 as an electron donor to reduce CO_2 for methane production and is widely found in AD systems of straw-based biomass [45]. *Norank_Bathyarchaeia* can metabolize a wide range of substances, including H_2, CO_2, CH_4, methanol, formic acid, and methylamine, to produce acetyl coenzyme A [43]. This will inevitably lead to substrate competition with hydrogenotrophic methanogens, thus inhibiting their growth and metabolism.

3.5. Prediction of microbial function

Functional annotation of microbial communities via FAPROTAX database predicted NBW-mediated modulation of microbial functional gene expression in AD systems (Fig. 7). Bacterial functional profiles revealed that H_2-NBW supplementation enhanced organic matter degradation pathways (aromatic compound degradation and cellulolysis) and nitrogen/sulfur cycling functions (Fig. 7 and Fig. S2). Robust N/S cycling maintains proton gradient equilibrium across cellular membranes, thereby sustaining system stability and metabolic efficiency [46]. This mechanistic alignment is evidenced by the documented capacity of Air-NBW to mitigate ammonia inhibition in ammonia-rich swine manure digestion [47]. Additionally, CO_2-NBW exhibited elevated expression of aromatic hydrocarbon degradation, ligninolysis, and xylanolysis genes. CH_4-NBW demonstrated strengthened capabilities in aromatic hydrocarbon degradation and chitinolysis activity. This parallels observations in rice straw AD where CO_2/O_2-NBW supplementation enhanced key enzymatic gene expression during hydrolysis-acidification, consequently improving substrate degradation rates [48].

Archaeal functional gene expression revealed gas-specific NBW modulation of methanogenic pathways (Fig. 8). Compared to the control, H_2-NBW and CO_2-NBW systems exhibited 32.66 % and 29.23 % reductions in acetoclastic methanogenesis genes, while hydrogenotrophic methanogenesis genes increased by 35.28 % and 51.34 %, respectively. This hydrogenotrophic enhancement stems from H_2/CO_2 serving as direct electron donors/substrates for hydrogenotrophic methanogens. Additionally, H_2-NBW upregulated formate-based methanogenesis genes. Conversely, the expression of the hydrogenotrophic

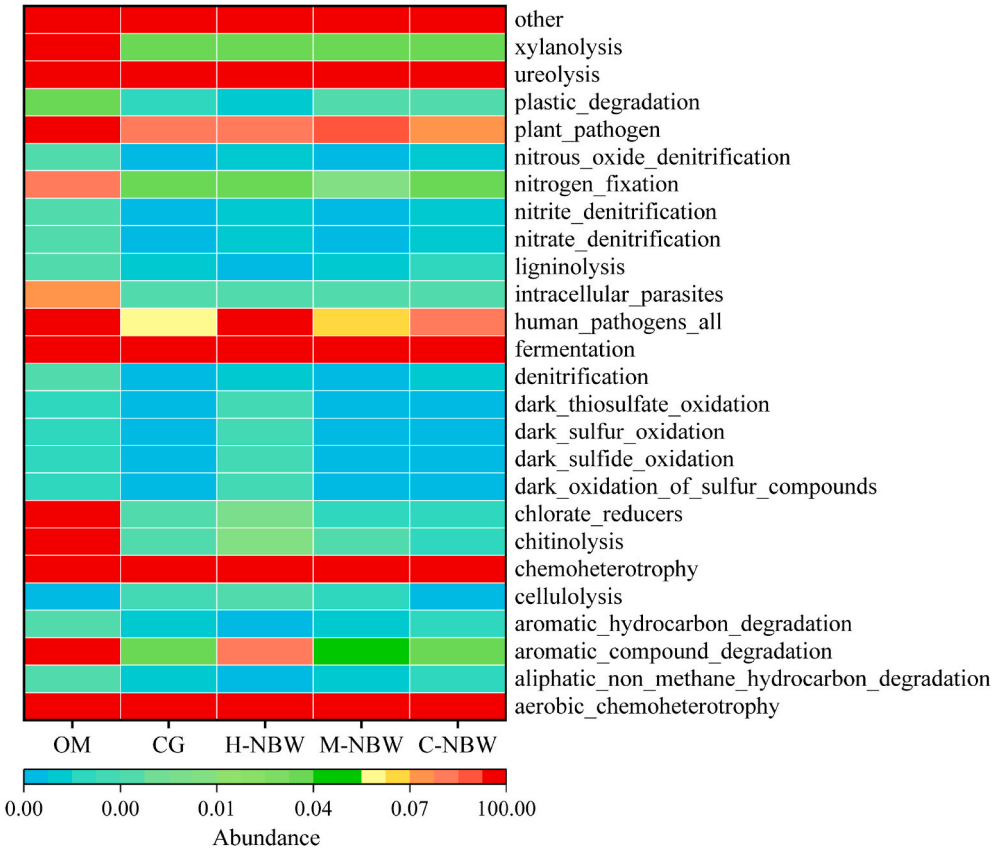


Fig. 7. Functional gene expression profiles of bacteria (OM: original microorganisms, CG: control group, H-NBW: H_2-NBW, M-NBW: CH_4-NBW, C-NBW: CO_2-NBW).

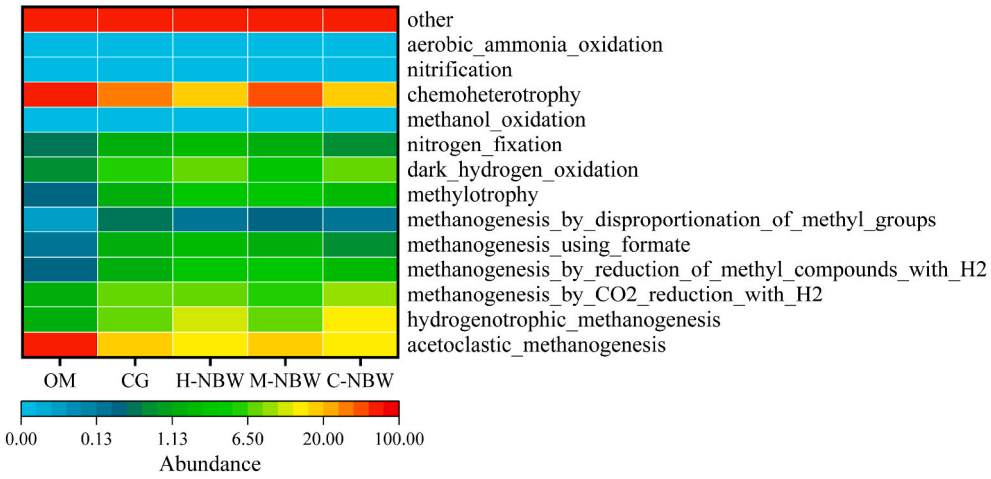


Fig. 8. Functional gene expression profiles of archaea (OM: original microorganisms, CG: control group, H-NBW: H_2-NBW, M-NBW: CH_4-NBW, C-NBW: CO_2-NBW).

methanogenesis function was decreased by 13.64 % in the CH_4-NBW group compared with the control group. Notably, the acetoclastic methanogenesis pathway of CH_4-NBW was 37.04 % and 30.41 % higher than that of H_2-NBW and CO_2-NBW groups, respectively. This phenomenon may be associated with the elevated relative abundance of *norank_Bathyarchaeia*. It has been demonstrated that *norank_Bathyarchaeia* is capable of metabolising H_2 and CO_2 to produce acetate, thereby inhibiting hydrogenotrophic methanogenesis while enhancing acetoclastic methanogenesis [43]. Consequently, H_2-NBW and CO_2-NBW promoted a metabolic shift from acetoclastic to hydrogenotrophic and methylotrophic methanogenic pathways. In contrast, CH_4-NBW exerted minimal effects on acetoclastic and hydrogenotrophic pathways but selectively enhanced methylotrophic methanogenesis.

3.6. Mechanism analysis

As illustrated in Fig. 9a, nuclear magnetic resonance (NMR) relaxation times were significantly prolonged in NBW compared to deionized water. The T_2 relaxation time measured 2713.19 ms for deionized water versus 2870.47 ms (H_2-NBW), 2804.70 ms (CH_4-NBW), and 2946.07 ms (CO_2-NBW). Extended relaxation times indicate enhanced water molecular mobility, which facilitates molecular diffusion and mass transfer processes [49]. In AD systems, the elevated mobility and diffusivity of NBW promote cellulose hydrolysis efficiency [27]. This hydrodynamic enhancement directly improves mass transfer kinetics, accelerating substrate flux and energy transduction [50]. Optimized mass transfer efficiency supports microbial metabolic activities, thereby increasing substrate utilization. In this research, NBs selectively enriched functional microbial consortia, stimulating substrate degradation and methanogenic gene expression. This leads to substantial compositional changes in the digestate, thereby affecting its physicochemical properties. These effects include decreased viscosity and surface tension, as well as increased specific heat capacity. Therefore, under conditions of enhanced thermal stability and flow characteristics, microbial activity, mass transfer, and NBs become more stable.

In summary, NBW operates through synergistic heat-mass transfer and microbial mechanisms. By establishing a "mass transfer-microorganisms-thermophysical properties" positive feedback loop (Fig. 9b), comprehensive system enhancement was achieved. This enhancement was realized through four interconnected mechanisms: (1) enhancement of mass transfer efficiency; (2) selective microbial enrichment and functional amplification; (3) alteration of material flux and spatial distribution; (4) improvement of digestate flow properties and thermal stability. At the microbial level, the efficient mass transfer facilitated by NBW inevitably alters the microbial community structure and enhances metabolic activity. This represents the core mechanism of NBW's effect. Concurrently, accelerated microbial metabolism improves substrate utilization efficiency, thereby altering substance concentrations. These differing substance distribution patterns lead to variations in the physicochemical properties of the digestate, which indirectly influence its thermophysical properties. Ultimately, this creates a more stable environment for both AD microorganisms and NBs, fostering a robust "mass transfer-microbial activity-thermophysical properties" cycle within the system. While the inherent physicochemical properties

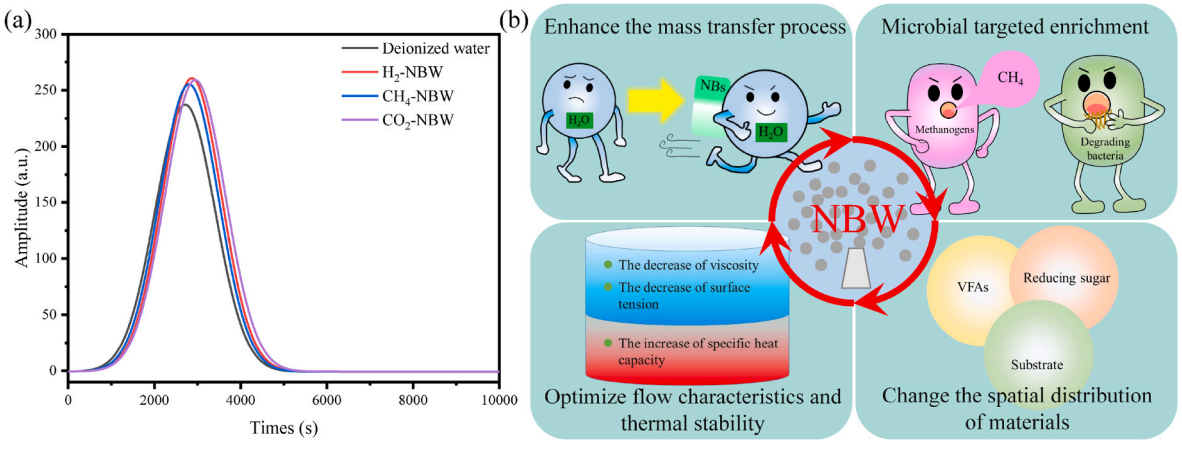


Fig. 9. (a) NMR T_2 relaxation times of different NBW types, (b) mechanistic analysis of NBW-enhanced processes.

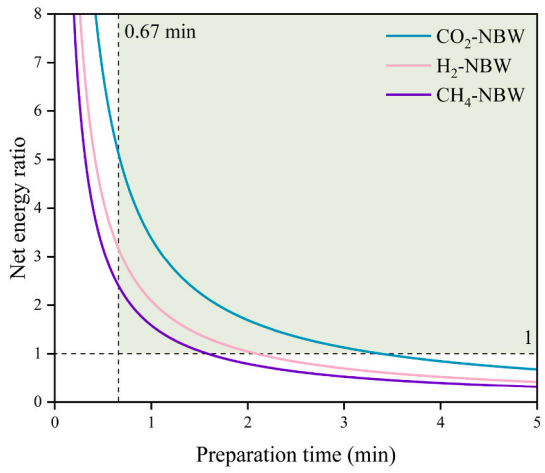


Fig. 10. NBW net energy ratio versus circulation preparation time (light green area indicates circulation preparation time for achieving net energy gain).

of NBW exert some direct influence on the digestate, the complex gas-liquid-solid system and dynamic flow conditions inherent to the digestate itself dilute the direct effects of the NBs. Consequently, the direct impact of NBW on the digestate is less significant than its indirect impact.

3.7. Energy efficiency analysis and outlook

The two primary sources of energy input involved in the application of NBs in AD are the electrical energy consumed by the operation of the nanobubble generator and the calorific value contained in the gas itself. However, due to limitations in current nanobubble detection techniques (such as nanoparticle interference and bubble stability), it is difficult to accurately determine the amount of gas dissolved in the liquid phase. Therefore, this paper only briefly discusses the energy efficiency relationship between the energy consumption for preparing NBW and the resulting energy gain. The calculation results show that under different conditions, the net energy ratios of the NBW groups are 0.42 (H_2-NBW), 0.32 (CH_4-NBW), and 0.68 (CO_2-NBW), respectively. As shown in Fig. 10, the net energy ratio is closely related to the circulation preparation time of NBW. As the circulation time increases, the net energy ratio gradually decreases. With a methane production rate increase of 15.67 %, the net energy ratio reaches 1 when the circulation preparation time of CO_2-NBW is 3.21 min. However, under conditions without circulation preparation (t = 0.67 min), the net energy ratios for adding NBW are 3.13 (H_2-NBW), 2.37 (CH_4-NBW), and 5.07 (CO_2-NBW), respectively. This indicates that in practical engineering applications, the circulation preparation time per unit volume of NBW should be minimized or circulation preparation should be avoided altogether to enhance the system's energy efficiency.

The improvement of thermophysical properties of digestate by NBW suggests its significant potential in reducing energy consumption associated with mixing and dynamic heating. Compared to traditional chemical additives, NBW does not require subsequent recovery and will not cause secondary pollution to the AD system, thereby further reducing its operational costs. Nevertheless, future studies should include a comprehensive life cycle analysis encompassing the gas production stage to comprehensively evaluate its overall benefits. For other organic wastes (such as food waste, sludge, or agricultural residues), nanobubble technology also holds promise for enhancing overall system efficiency by improving mass and heat transfer and regulating microbial metabolism. It is particularly noteworthy that the NB-AD technology can be efficiently integrated with other bioenergy systems. For instance, it can be combined with biohydrogen production systems to convert a low concentration of hydrogen into NBW for enhancing the AD process, thereby improving overall energy recovery efficiency while reducing operational costs. Although the integration of nanobubble technology into existing biogas projects demonstrates good operational feasibility, its large-scale application still faces numerous challenges, including optimal dosing strategies and uniform distribution during continuous operation. Future research should focus on long-term performance evaluations at pilot scale, techno-economic analyses of the full life cycle, and detailed investigations into the interaction mechanisms between NBs and complex microbial communities.

4. Conclusion

This research evaluated the effects of NBW on semi-continuous AD, elucidating its enhancement mechanisms through thermophysical and microbial analyses. NBW demonstrated sustained performance improvement in corn stover digestion, with mean daily methane production during stable operation increasing by 9.68 ± 2.08 % (H_2-NBW), 7.34 ± 2.07 % (CH_4-NBW), and 15.67 ± 1.96 % (CO_2-NBW) relative to the control. On one hand, NBW enhanced substrate degradation efficiency, leading to improved removal rates of TS and VS. On the other hand, NBW accelerated the consumption of reducing sugars and SCOD while enhancing the VFAs generation rate. Concurrently, NBW effectively improved the thermophysical properties of the digestate, thereby optimizing heat and mass transfer processes. Compared to the control, H_2-NBW increased the specific heat capacity by 1.23 ± 0.33 % and decreased viscosity and surface tension by 11.13 ± 0.20 % and 2.56 ± 1.40 %, respectively. Furthermore, NBW enriched the microbial community diversity within the AD system, increasing the relative abundance of various hydrolytic/acidogenic bacteria, as well as diverse methanogenic archaea. Both H_2-NBW and CO_2-NBW induced a shift in the methanogenic pathway towards hydrogenotrophic methanogenesis. However, CH_4-NBW led to the significant enrichment of *norank_Bathyarchaeia*, which consequently suppressed the functional expression of hydrogenotrophic methanogenesis. Collectively, these findings indicate that NBs operate through a synergistic mechanism involving enhanced heat-mass transfer and microbial community modulation, ultimately improving methane production efficiency and system stability.

CRediT authorship contribution statement

Yongkang Yuan: Writing – original draft, Investigation, Data curation, Conceptualization. **Tingting Hou:** Writing – review & editing, Methodology, Conceptualization. **Pengfei Li:** Writing – review & editing. **Hui Sun:** Writing – review & editing. **Youzhou Jiao:** Writing – review & editing. **Gang Li:** Writing – review & editing. **Chao He:** Writing – original draft, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2025.124921.

Data availability

Data will be made available on request.

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